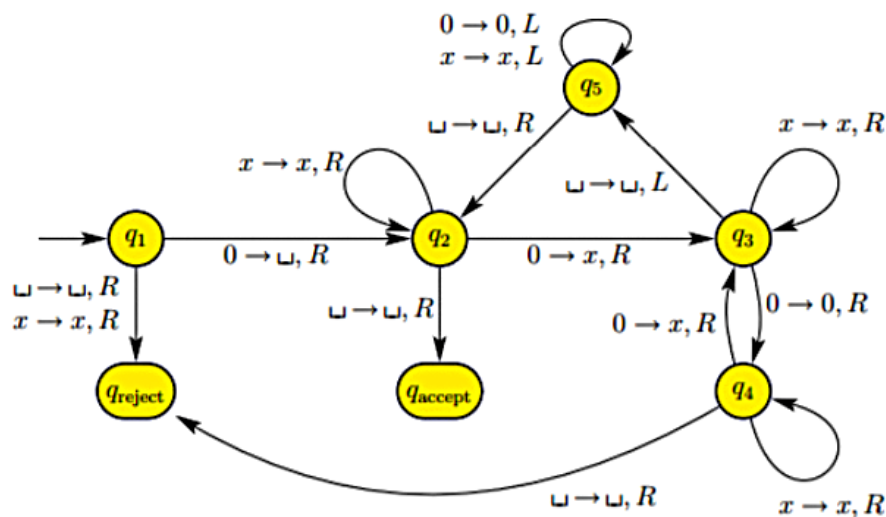


WORKSHOP

TURING MACHINES AND DECIDABILITY

- Given the language

$$L = \{ w \mid w \in \Sigma^*, \Sigma = \{ a, b \}, w \text{ does not contain the substring } aba \}$$
 - Construct a Turing Machine that accepts L
 - Trace the execution for the input $bababb$
 - Show that L is decidable
- Given the language $L = \{ ww \mid w \in \{a, b\}^+ \}$
 - Design a Turing machine that accepts L
 - Describe how the designed TM works
 - Trace the execution for the input $abab$
 - Trace the execution for the input bab
- Design a Turing machine with no more than 5 states that accepts the language $(01 + 011)^*$
- Given the language $L = \{ w \in \{a, b\}^* \mid \#a(w) \leq 2\#b(w) \}$
 - Design a Turing machine that accepts L
 - Is L decidable? Justify your answer
- Design a Turing Machine that adds one to a given binary number. The input binary number is written on the tape, surrounded by blank symbols. The machine must halt in an accept state with the incremented binary number left on the tape.
- The Turing Machine showed in the next figure recognizes the language $L = \{ 0^{2^n} \mid n \geq 0 \}$



- a) Analyze how the machine processes the strings:

0000000

000000

00000

- b) Explain in words how the machine works (without describing each individual transition, explain the general idea, how the machine determines whether the length of the string is a power of 2)

7. Given the language $L = \{ w\#x \mid w \text{ is a substring of } x \}$

a) Design a Turing machine that accepts L

b) Describe how the designed TM works

8. Given the language $L = \{ x+y=z \mid x,y,z \text{ are binary integers and } x+y=z \}$

a) Design a Turing machine that accepts L

b) Describe how the designed TM works